The components on which these contributions rest, namely the causal plant estimator, the bounded adaptation path and the eight-gate evaluation harness, are described where they are used. They are technical enablers rather than separate claims.

1.5 Thesis Structure

This thesis investigates how hydraulic actuator dynamics learned from high-fidelity simulation can be reduced, transferred into a computationally efficient robotic simulation and used for reinforcement-learning control, grounding its investigation in the identifiability of the actuator models and in the robustness of the controllers built on them. The document is organized as follows:

Chapter 2 establishes the background against which the remainder is read. It introduces the machine class and its electro-hydraulic actuation, recalls the kinematic tools underlying the classical controller, and reviews how conventional model-based control handles hydraulic uncertainty. It then treats the learning methods this work builds on, namely data-driven dynamics models and the pitfalls of identifying them from closed-loop data, reduced-order and grey-box modelling, domain randomisation and latent adaptation, and policy optimisation with structured priors. The chapter closes by stating the gap that remains and the position of the three control stacks examined here.

Chapter 3 analyses the platform, using the MULAG FME 500 mounted on a Unimog carrier as the case study. It describes the rigid bodies and the serial chain they form, the coordinate mappings between task space and cylinder space, and the closed kinematic loops introduced by the cylinder mounts. It specifies what the controller can observe, separating measured quantities from quantities produced by a model, and presents the high-fidelity Simscape model from which the actuator data derives. The chapter also states the task, the plant capability limits and the acceptance criteria against which every later result is scored.

Chapter 4 presents the data generation strategy and answers the first research question. It describes the excitation design, the randomised factor set defining the family of plants and the procedures that keep training, validation and reserved test configurations separate. It then reports the identifiability audit applied to the resulting collection, including the measured finding that excitation gathered under a task controller does not separate plants and the open-loop collection that repairs this.

Chapter 5 develops the learned models and answers the first three research questions. It presents the inverse model that maps a desired cylinder motion to a valve command, the forward model that predicts the response, and the evidence that supervised accuracy alone does not establish that either model retains command authority. It then presents the command-conditioned reduced-order model that serves as the deployment plant inside GPU-parallel training. The chapter closes with the material required for the final research question, namely the privileged latent and its measured limits, the causal estimator that infers a deployable substitute from observable history, and the bounded adapter through which that estimate reaches the actuator path.

Chapter 6 describes the control framework and its integration with the simulator, and answers the fourth research question. It presents the damped least-squares teacher used for demonstration and data collection, the observation and action interfaces shared by all three